

Title: Improving Survival Prediction in NSCLC through Clinical Report Embeddings: A Comparison with Structured Tabular Data and Multimodal Fusion Model.

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Background

Accurate prediction of short-term survival in non-small cell lung cancer (NSCLC) is challenging, especially at the start of immunotherapy. Structured data often lacks the clinical context of medical reports. We evaluated whether embeddings from clinical reports, generated using large language models (LLMs), improve the prediction of 6-month survival after immunotherapy, and whether multimodal integration provides further benefit.

Methods

We analyzed 647 NSCLC patients from the Apollo 11 study (NCT05550961) at Fondazione IRCCS Istituto Nazionale dei Tumori, Milan. Clinical reports were processed with Qwen2.5-7B-Instruct-1M to generate 3584-dimensional embeddings. Data were split into training (85%) and test (15%) sets. Three models were built: (i) embeddings, (ii) tabular data, (iii) multimodal late-fusion. Pipelines included preprocessing, dimensionality reduction, and logistic regression. Performance was assessed by ROC AUC using the DeLong method.

Results

Model	Train AUC	CV AUC	Test AUC
Real world model from tabular data	0.74 ± 0.04	0.74 ± 0.04	0.71 ± 0.1
LLM-based text based	0.81 ± 0.04	0.71 ± 0.04	0.77 ± 0.1
Late-fusion Hybrid (both RWD+Text)	0.82 ± 0.04	0.77 ± 0.04	0.78 ± 0.09

On the test set, the embedding-based model outperformed the tabular baseline (AUC 0.77 vs 0.71). Cross-validation showed lower stability for embeddings (CV AUC 0.71). The multimodal model achieved the best performance (test AUC 0.78; CV AUC 0.77), significantly outperforming the baseline (bootstrap $p=0.0107$).

Conclusions

Embeddings extracted from EHR clinical reports effectively predicted 6-month survival in NSCLC patients treated with immunotherapy. As a standalone approach, the LLM-based model achieved strong performance compared to tabular data, reducing clinician workload without loss of accuracy. The hybrid approach further improved prediction, indicating that embeddings capture complementary, clinically relevant information. These findings highlight the value of automatic extraction tools for predictive modelling.